EXP.NO: 4	Create a web form that accepts a user's name and age. Write a Servlet to process the form data and display it back on the browser.
DATE:26/02/2025	

AIM:

To create a simple web application that includes a form to accept a user's name and age, and then use a Servlet to process the submitted data. The Servlet will retrieve the form data, process it, and display the values back on the browser.

ALGORITHM:

1. **Input:**

 The user fills out a web form with their **name** and **age** and submits it.

2. **Form Handling and Servlet Processing:**

1. **Create the Web Form:**
   - Create an HTML form with **two input fields**: one for the user's **name** and one for the user's **age**.
   - Add a **submit button** to the form.
   - Set the form's action attribute to the URL of the **Servlet** and the method to POST (or GET).
2. **Create the Servlet Class:**
   - Write a Servlet that processes the **form data** submitted by the user.
   - The Servlet will override the doPost() (or doGet() depending on the form method) to handle form submissions.
   - The Servlet will **retrieve the name and age** parameters from the request object using request.getParameter("name") and request.getParameter("age").

3. **Process the Data in the Servlet:**
    - Extract the name and age values submitted from the form.
    - **Validate the input** (e.g., check if the age is a valid number).
    - Prepare a **response** that will display the user's **name** and **age** back on the browser.
4. **Send Response to the Browser:**
    - The Servlet will send a **response** to the browser containing a message like:
    - "Hello, [name]! You are [age] years old."
    - The response will be displayed to the user in the browser.
5. **Deploy the Servlet:**
    - Compile the Servlet and deploy it to a **Servlet container** (like **Apache Tomcat**).
    - Ensure the web.xml file or Servlet annotations are used to map the Servlet to the URL defined in the form's action attribute.
6. **Test the Application:**
    - Open the web browser, fill in the form with the name and age, and submit it.
    - The Servlet will process the data and display the message back to the user in the browser.

Program:

```
<%@ page import="java.io.*" %>
<!DOCTYPE html>
<html>
<head>
<title>User Form</title>
<style>
body {
   font-family: Arial;
   background: #111;
   color: white;
```

```
}
.container {
    width: 400px;
    margin: 80px auto;
    background: #222;
    padding: 20px;
    border-radius: 10px;
}
h2 {
    text-align: center;
    color: #ff4d4d;
}
input {
    width: 100%;
    padding: 8px;
    margin: 8px 0;
}
button {
    width: 100%;
    padding: 10px;
    background: #ff4d4d;
    color: white;
    border: none;
}
.result {
    margin-top: 20px;
    text-align: center;
}
</style>
</head>

<body>

<div class="container">
    <h2>User Form</h2>

    <form method="post">
        <input type="text" name="name" placeholder="Enter Name"
required>
        <input type="number" name="age" placeholder="Enter Age"
required>
        <button type="submit">Submit</button>
    </form>
```

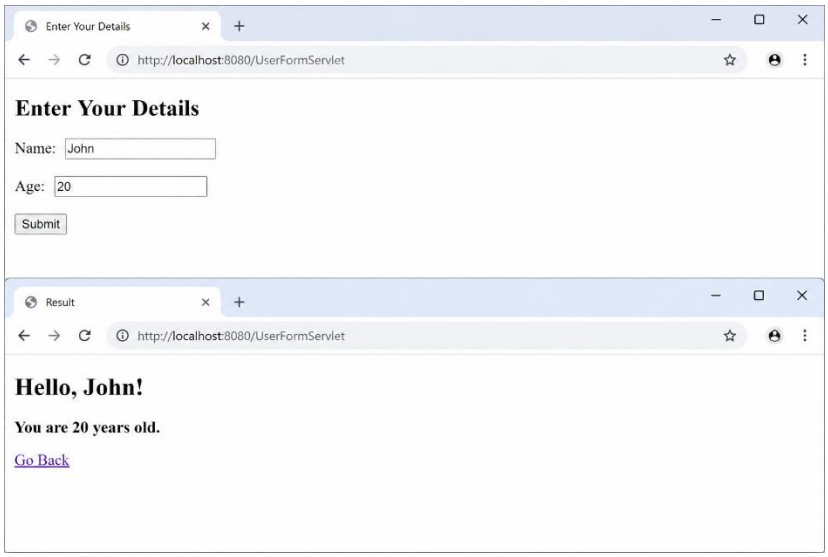
```
    <div class="result">
        <%
            String name = request.getParameter("name");
            String age = request.getParameter("age");

            if(name != null && age != null) {
                try {
                    int a = Integer.parseInt(age);
                    out.println("<p>Hello, " + name + "! You are " + a + " years
old.</p>");
                } catch(Exception e) {
                    out.println("<p>Invalid age entered.</p>");
                }
            }
        %>
    </div>

</div>

</body>
</html>
```

Output:



RESULT:

A Servlet is created to process the HTML form data and display it back on the browser.